INCH-POUND

A-A-54892
23 July 1993

COMMERCIAL ITEM DESCRIPTION

TRIMMER, DENTAL MODEL

WHEEL, ABRASIVE, SILICON CARBIDE, DENTAL MODEL, TRIMMING

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-T-37538A, which is cancelled.

This Commercial Item Description covers a dental model trimmer and replacement wheel suitable for dental laboratory use.

Salient characteristics:

Dental Model Trimmer. The dental model trimmer shall be furnished complete with a grinding wheel, a 4 foot (minimum length) water inlet tube, a faucet adapter, a 2-1/2 foot plastic hose, a 4 foot length of copper tubing with a tap-in-tap connector and a shut-off valve. It shall be in accordance with figure 1.

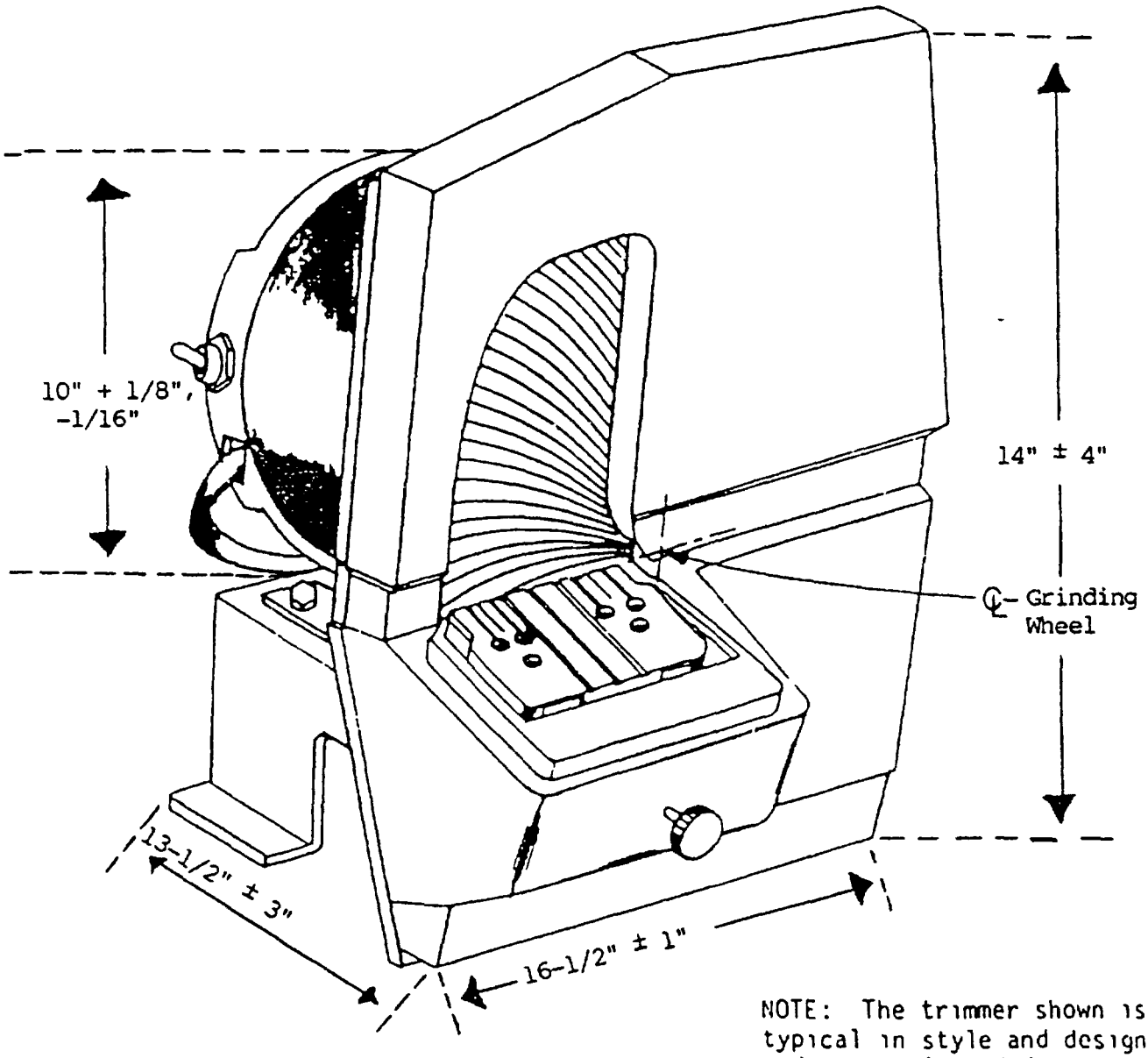
The trimmer shall be heavy duty, electrically-operated, and suitable for trimming stone and plaster models in the dental laboratory. The trimmer shall consist of a motor driven grinding wheel with an aluminum back-up plate and a water spray system for wetting the grinding wheel and removing debris from the wheel during operation. The motor shall be dynamically balanced, 1/3 horsepower, 1725 revolutions per minute, single phase, 115 volts, 60 Hertz, capacitor start or split phase. The motor shall be equipped with a heavy duty electrical power cable assembly and a suitable power switch for energizing and de-energizing the motor.

The trimmer shall be equipped with a slotted or perforated removable aluminum tray.

AMSC N/A FSC 6520

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NOTE: The trimmer shown is typical in style and design and is not intended to restrict or preclude the purchase of trimmers which are otherwise in accordance with the requirements of this document.

FIGURE 1. Dental model trimmer.

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The power cable assembly shall be a 3-conductor cord, not less than 5 feet long with a 3-prong plug attached to one end. The plug shall be rated as suitable for use with a 115V, 15A, 2 pole, 3-wire, grounding type receptacle.

The grinding wheel housing shall be of two-piece construction, cast aluminum, with the front housing hinged or removable with thumb screws, for easy access to the grinding wheel. The front and rear housing shall be sealed with a rubber gasket to provide a water-tight housing when closed.

The front wheel housing shall have an opening when the work tray is in the horizontal position, which precludes horizontal motion of the model. The work tray shall be cast aluminum, adjustable, and have the following dimensions: 5 plus or minus 1/2 inches wide by 4-3/4, plus 1/2, minus 1/2 inches deep and a minimum thickness of 5/32 inch. The work tray shall be provided with holes or slots. Beneath the work tray shall be a drip tray which empties into a waste reservoir at the bottom of the grinding wheel housing.

The trimmer shall be provided with a spray system for wetting the grinding wheel and washing away debris. The spray system shall be so designed that the water spreads evenly over the full radius of the grinding wheel. The grinding wheel housing shall be equipped with a water inlet fitting and a drain hose connection or drain pipe for carrying away grinding debris. The drain outlet shall have a minimum inside diameter of 3/4 inch.

The water inlet tube supplied with the trimmer shall have a minimum length of 4 feet and a faucet adapter for attaching to unthreaded faucets. The flexible rubber or vinyl translucent plastic hose supplied with the trimmer shall have a minimum length of 2-1/2 feet. The copper tubing supplied with the trimmer shall be suitable for permanent connection to water pipe. Any tools necessary to change the grinding wheel shall be supplied with the trimmer. All fittings, tubes and hoses shall properly fit, and function with, their intended mating components. The trimmer unit shall include an aluminum back-up plate to support the wheel while in operation.

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Dental Model Trimming Wheel. The trimming wheel of the dental trimmer unit shall be connected directly to the shaft of the motor. The trimming wheel shall be a reversible, medium grain size, silicon carbide, resinoid bond abrasive wheel. The grinding wheel shall be 10 plus 1/8, minus 1/16 inches in diameter, and 1/4 plus 1/64, minus 0 inch thick with a 1 plus 0.002, minus 0.005 inch bore. The grinding surfaces on both sides of the wheel shall have spiral grooves to prevent excessive loading. The molded spirals shall run in a counterclockwise direction away from the center, as viewed from either side of the wheel. There shall be 12 spirals per inch at the wheel periphery. Grit size shall be 46. Maximum runout on the face and sides of the wheel shall be plus or minus 0.010 inch. Wheel sides shall not be concave by more than 0.010 inch. Each wheel shall be statically balanced to 1 gram or less.

Mounting provisions for the complete trimmer. The trimmer shall be provided with a minimum of four mounting holes, each of sufficient size to accept a minimum bolt diameter of 1/4 inch. As an alternate, the trimmer shall be provided with four rubber caster cups suitable for use under the base to prevent slipping.

Acoustical noise levels for the complete trimmer. The trimmer shall not produce noise levels above 85 dBA at the operator's position.

Underwriters' Laboratories approval. The complete trimmer shall be in accordance with Underwriters' Laboratories (UL) 1262, Standard for Laboratory Equipment.

Identification marking. Each unit shall be identified with the name or registered trademark of the manufacturer, model designation, serial number, stock number, and contract or purchase order number, horsepower, number of phases, current, voltage, speed, and cycles. Markings shall be provided on one or two readily visible nameplates, permanently affixed. The markings on the plates shall be legible and permanent and shall be accomplished by stamping. Each wheel shall be marked in a legible manner with the manufacturer's name or trademark, diameter, thickness, bore, grain (grit) size and maximum operating speed.

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Workmanship. The model trimmer and abrasive wheel shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One trimmer along with accessories and service data or one abrasive wheel, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, at inspection level II with an AQL of 2.5 (percent defective) except metal surface, welds, dimensions and identification marking shall have an AQL of 4.0 (percent defective). The unit of product for examination purposes shall be one trimmer and one wheel, as applicable.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The unit of product for test purposes shall be one complete trimmer and one replacement wheel, as applicable. Sampling for tests shall be conducted at an inspection level of 100 percent for performance testing or one trimmer for noise testing.

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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

Performance test. The trimmer shall be set up on a suitable bench top location with water supply hose and drain hose connected for operation. Connect electric plug to 115 volt, 60 Hz receptacle. Turn on water faucet to a slight opening. Turn on trimmer. Using a piece of plaster of approximate size of a model to be trimmed, proceed to grind the plaster for 2 to 3 minutes. After grinding operation, allow water and motor to remain operating for 15 seconds to wash away plaster sludge. Inspect for water leakage around grinding wheel housing gasket. There shall be no leakage. Inspect for vibration. There shall be no vibration with grinding wheel in operation. Turn off water supply and electric power. Open front housing cover and inspect grinding wheel for clogging by plaster. There shall be no clogging of the grinding wheel. Excessive plaster sludge shall have been washed from the reservoir.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document.

National Stock Number		Item Identification
6520-00-564-4050	-	TRIMMER, DENTAL MODEL 115V, 60 Hz
6520-00-564-4055	-	WHEEL, ABRASIVE, SILICON CARBIDE

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MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6520-2540

Location: TRIMR14C.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54892	2. DOCUMENT DATE (YYMMDD): 23 July 1993

3. DOCUMENT TITLE: TRIMMER, DENTAL MODEL
WHEEL, ABRASIVE, SILICON CARBIDE, DENTAL MODEL, TRIMMING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME: DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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